MERCYLIN MUTHONI

AI/ML Engineer

CONTACT

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Mercylin Muthoni

MercylinM

Portfolio

EDUCATION

AkiraChix
codeHive - Diploma in Information Technology
February 2025 - present

SKILLS

Languages: Python, SQL, JavaScript
Libraries & Frameworks: Pandas, NumPy, scikit-learn, TensorFlow, PyTorch, HuggingFace Transformers, OpenCV, Matplotlib, Seaborn, Plotly, FastAPI, Django
Databases & Vector Search: PostgreSQL, ChromaDB, Supabase
Platforms & Tools: Jupyter Notebook/Lab, Google Colab, Google Cloud, GitHub, Postman, Cloud Deployment

REFERENCE

Linda Kamau
Founder & Executive Director
AkiraChix
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Caroline Mutua
Product Manager
Microsoft
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PERSONAL STATEMENT

Will the trade-off between privacy and convenience become the norm in our digitised future? Or will people reclaim control, heralding a new era for the internet? As public concerns about online privacy intensify and cultural movements like social media-free living gain momentum, Mercylin believes that privacy must be a universal right, not a privilege reserved for the few. As a software engineer, she hopes to contribute to privacy-first technologies that ensure privacy does not come with a price tag, accessible only to those who can afford it.

PROJECTS

Recos - Recruitment Intelligence System(Agentic AI)
Recos is an Agentic AI that automates candidate evaluation and interview analysis by assessing candidate profiles, performing real-time sentiment analysis during interviews, and generating data-driven hiring recommendations, ultimately reducing skill mismatches and improving hiring accuracy.
Approach & Key Learnings:

- Integrated Odoo recruitment data with PostgreSQL and built a sync pipeline to improve candidate data accuracy and availability.
- Experimented with AssemblyAI for live transcription in a Google Meet addon with interview analytics, then switched to Google Cloud Speech-to-Text for better accuracy and real-time performance.
- Developed an NLP pipeline using Gemini LLM for skill extraction and semantic matching from resumes, replacing the earlier Amjad Skill Extractor for more reliable and context-aware results.
- Created an interview analysis pipeline that performs sentiment analysis, generates follow-up questions, and produces candidate performance reports with hiring recommendations. Streamlined the system by consolidating into a Gemini-powered backend.
- Optimized prompt engineering for Gemini to ensure consistent skill extraction and sentiment analysis across varied profiles, crucial after transitioning from rule-based to LLM-powered methods.

Movie Recommendation System (RAG + AI Agent)
Developed a personalised movie recommendation system that suggests movies based on user preferences and context. This system improved the quality and trustworthiness of recommendations by providing clear and relevant suggestions tailored to individual users.
Approach & Key Learnings:

- Leveraged ChromaDB to embed and index movie metadata, enabling fast and accurate semantic search for content retrieval.
- Orchestrated a modular recommendation workflow using Google ADK, facilitating flexible and scalable reasoning through composable agent steps.
- Experimented with multiple prompting strategies, including few-shot, zero-shot, and chain-of-thought; identified a hybrid zero-shot and chain-of-thought approach as the most reliable and efficient.
- Gained insights into the critical role of prompt engineering combined with modular agent architecture for building scalable, maintainable recommendation systems.

Breast Cancer Image Classification (Computer Vision)
Developed an AI model to classify breast tissue images, achieving 75% overall accuracy and demonstrating the feasibility of AI-assisted pathology for early cancer detection.
Approach & Key Learnings:

- Applied preprocessing techniques, including resizing, normalization, and augmentation, to address class imbalance and enhance model generalization.
- Utilized EfficientNet with transfer learning, balancing computational efficiency and predictive accuracy, tailored to the specific dataset characteristics.
- Evaluated model performance using precision, recall, and F1-score metrics at the class level, ensuring balanced results across categories critical for healthcare applications.
- Emphasized the importance of robust data preprocessing and metric-driven evaluation over solely focusing on advanced architectures or interpretability, recognizing that balanced and reliable performance across classes is crucial in medical AI solutions.